

UNIVERSITY OF LISBON

ISEG- LISBON SCHOOL OF ECONOMICS AND MANAGEMENT

EXAM - JANUARY 2022

ADVANCED ECONOMETRICS

Module Convenor:
Paulo M.D.C. Parente

Use answer booklets of ISEG (folhas de teste do ISEG)

Instructions (please read before starting): Write in a clear legible manner in ink/ballpoint. Do not use pencils or erasable pens. Approved calculators are permitted. Only one sheet (2 sides A4) of notes made exclusively by the student may be consulted (no material distributed by the teacher in any form is allowed). Whenever conducting a test use a 5% significance level unless stated otherwise. Also be sure to state null and alternative hypotheses, null distribution (with degrees of freedom), rejection criterion (critical values and rejection region) and outcome. If you are asked to derive something, give all intermediate steps also. Do not answer questions with a “yes” or “no” only, but carefully justify your answer.

Section A - Topics in Microeconometrics

Question 1

1. Consider a binary random variable Y_i such that

$$P(Y_i = 1|D_i) = G(\alpha + \beta D_i), \quad i = 1, \dots, n,$$

where D_i is a dummy variable and $G(\cdot)$ is a strictly increasing known function that satisfies $0 < G(\cdot) < 1$. Assume that the sample $\{(Y_i, D_i)\}_{i=1}^n$ corresponds to a random sample and denote n_{ij} the number of elements of the sample that satisfy $Y = i$ and $D = j$, where $i = 0, 1$ and $j = 0, 1$. Assume initially that n_{ij} is strictly positive for all i and j .

- (a) (2 marks) Show that the maximum likelihood estimators of α and β are given by

$$\begin{aligned}\hat{\alpha} &= G^{-1}\left(\frac{n_{10}}{n_{10} + n_{00}}\right), \\ \hat{\beta} &= G^{-1}\left(\frac{n_{11}}{n_{01} + n_{11}}\right) - G^{-1}\left(\frac{n_{10}}{n_{10} + n_{00}}\right),\end{aligned}$$

where $G^{-1}(\cdot)$ is the inverse function of $G(\cdot)$ [**Hint:** Maximize the log-likelihood function with respect to $A = G(\alpha)$ and $B = G(\alpha + \beta)$ and use the invariance of maximum likelihood estimators to transformations to obtain $\hat{\alpha}$ and $\hat{\beta}$.]

- (b) (2 marks) Show that the log-likelihood function evaluated at $\hat{\alpha}$ and $\hat{\beta}$ does not depend on $G(\cdot)$.
- (c) (2 marks) Prove that the maximum likelihood estimator does not exist if $n_{10} = 0$.
- (d) The data consist of 100 observations as follows:

		Y	
		0	1
D	0	24	28
	1	32	16

and assume that $G(z) = \exp(z) / [1 + \exp(z)]$.

- (1.5 marks) Obtain the maximum likelihood estimates of α and β .
- (2.5 marks) Test the hypothesis that $\beta = 0$ by using a likelihood ratio test.

Question 2

(2.5 marks) Assume that the distribution of Y^* conditional on the vector of exogenous regressors X is Poisson with parameter $\lambda = \exp(X'\beta)$. that is

$$\mathcal{P}(Y^* = y^* | X) = \frac{\lambda^{y^*} e^{-\lambda}}{y^*!}, y^* = 0, 1, 2, \dots$$

However, Y^* is censored, and we only observe Y defined as:

$$Y = \begin{cases} Y^* & \text{if } Y^* > 3 \\ 3 & \text{if } Y^* \leq 3 \end{cases}.$$

Assume additionally that a random sample is available $\{(Y_i, X_i')'\}_{i=1}^n$. Write down the log-likelihood function for this problem.

Section B - Topics in Time Series**Question 3**

(2.5 marks) Consider the ARMA(1,2) process

$$Y_t = 0.25Y_{t-1} + \varepsilon_t - 0.05\varepsilon_{t-1} - 0.05\varepsilon_{t-2},$$

where ε_t is a white noise process with variance σ_ε^2 . Show that this process is equivalent to a MA(1) process. [**Hint-** You may use the following result: Let $P_J(x)$ be a polynomial of order J in x , that is $P_J(x) = \sum_{j=0}^J a_j x^{J-j}$, where $a_0 \neq 0$, and the coefficients a_j , $j = 0, \dots, J$, are real or complex numbers, then $P_J(x) = a_0 \prod_{i=1}^J (x - r_i)$, where r_i , $i = 1, \dots, J$ are the real or complex roots of $P_J(x)$, with some of them possibly repeated]

Question 4

Consider the VAR(1) model $z_t = \Phi_1 z_{t-1} + \varepsilon_t$ where $z_t = (z_{1t}, z_{2t})'$ and ε_t is a 2×1 vector of white noise processes with

$$\text{var}(\varepsilon_t) = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}.$$

(a) (2 marks) Let

$$\Phi_1 = \begin{bmatrix} 1 & -0.6 \\ 0.5 & -0.7 \end{bmatrix}.$$

Obtain the roots of the characteristic equation and show that the process is stationary.

(b) (2 marks) Obtain the values of the elements of the matrices Ψ_ℓ , for $\ell = 0, 1, 2$ in the infinite moving average representation $z_t = \sum_{\ell=0}^{\infty} \Psi_\ell \varepsilon_{t-\ell}$.

(c) (1 marks) Obtain the impulse response function for z_{1t} to a shock to the variable z_{1t} of size σ_1 , for horizons $\ell = 0, 1, 2$.

[END OF PAPER]